

This assignment will not be graded and is only for practice.

Level 1

1) Consider two vectors $a = (1, -2, 2)$, $b = (4, 0, -3)$. Using the standard dot product **1 point** as the inner product, the projection of a in the direction of b is given by

- ☐ $(\frac{-8}{25}, 0, \frac{6}{25})$
- ☐ $(\frac{-8}{25}, \frac{6}{25})$
- ☐ $(\frac{8}{25}, 0, \frac{-6}{25})$
- ☐ $(\frac{8}{25}, \frac{-6}{25})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$(\frac{-8}{25}, 0, \frac{6}{25})$$

2) Consider an orthonormal basis $\{\frac{1}{\sqrt{5}}(1, 2), \frac{1}{\sqrt{5}}(-2, 1)\}$ of $\mathbb{R}^2$. If $(3, 5) = c_1 v_1 + c_2 v_2$ where $c_1, c_2 \in \mathbb{R}$ and $v_1 = \frac{1}{\sqrt{5}}(1, 2)$, $v_2 = \frac{1}{\sqrt{5}}(-2, 1)$, then choose the correct option. **1 point**

- ☐ $c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$
- ☐ $c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{1}{\sqrt{5}}$
- ☐ $c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$
- ☐ $c_1 = \frac{-13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$$c_1 = \frac{13}{\sqrt{5}}, c_2 = \frac{-1}{\sqrt{5}}$$

3) Consider an orthonormal basis $\{(1, 0, 0), (0, 1, 0)\}$ for a subspace W in $\mathbb{R}^3$. If $x = (1, 2, 3)$ is a vector in $\mathbb{R}^3$, then which of the following represents a vector in W whose distance from x is the least? Consider dot product as the standard inner product. **1 point**

☐ (2, 4, 0)

☐ (3, 4, 0)

☐ (4, 5, 0)

☐ (1, 2, 0)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1, 2, 0)

Level 2

Consider the inner product $\langle a, b \rangle = a_1b_1 - a_1b_2 - a_2b_1 + 4a_2b_2$ where $a = (a_1, a_2)$, $b = (b_1, b_2)$ are vectors in $\mathbb{R}^2$. Use this information to answer questions 4 and 5.

4) Let $x = (1, 2)$. Find the projection of x in the direction of $(3, 4)$ using the inner product defined above.

1 point

☐ $\frac{25}{49}(3, 4)$

☐ $\frac{11}{25}(3, 4)$

☐ $\frac{25}{49}(1, 2)$

☐ $\frac{11}{25}(1, 2)$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\frac{25}{49}(3, 4)$

5) Let $x = (1, 2)$, find the projection of x in a direction perpendicular to $(3, 4)$.

1 point

☐ $\frac{25}{49}(-4, 3)$

☐ $\frac{11}{25}(4, -3)$

☐ $(-\frac{26}{49}, -\frac{2}{49})$

☐ $(\frac{26}{49}, -\frac{2}{49})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(-\frac{26}{49}, -\frac{2}{49})$

6) Consider an orthogonal basis $\{(1, 2, 1), (-2, 0, 2)\}$ of a subspace W , of the inner **1 point** product space $\mathbb{R}^3$ with respect to the dot product. If $y = (1, 2, 3) \in \mathbb{R}^3$, then find $Proj_W(y)$.

☐ $(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})$

☐ $(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, \frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$

☐ $(\frac{1}{3}, -\frac{8}{3}, \frac{7}{3})$

☐ $(\frac{18}{\sqrt{6}} - \frac{32}{\sqrt{8}}, \frac{36}{\sqrt{8}}, -\frac{18}{\sqrt{6}} + \frac{32}{\sqrt{8}})$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$(\frac{1}{3}, \frac{8}{3}, \frac{7}{3})$

Your score is: 0/6